

Beyond Accuracy: A Climate-Aware Robustness Evaluation and Decision Support Framework for Crop Recommendation Systems

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Abstract—Crop recommendation systems built on classical machine learning have reached near-perfect accuracy on standard benchmarks, yet they fall short on a more consequential measure: they offer no insight into how their outputs hold up when environmental conditions shift. A model that confidently recommends rice under today’s climate but has never been stress-tested against a 2°C temperature rise or a 10% drop in rainfall is of limited value for real agricultural planning. The framework presented in this paper directly confronts that shortcoming. Rather than proposing yet another classifier, it wraps a standard high-accuracy backbone with four purpose-built components that move the system from static prediction toward genuine decision support: (1) a climate perturbation module that subjects the trained model to parameterized environmental stress scenarios; (2) a transition matrix analysis that tracks exactly which crops are displaced and by what substitutes under each scenario; (3) a heuristic counterfactual engine that translates a desired crop change into minimal, actionable soil-amendment steps; and (4) a risk-weighted portfolio recommender that replaces a single predicted label with a probability-ranked shortlist of viable alternatives. Additional layers—knowledge distillation for readable decision rules, SHAP-based attribution, adversarial micro-climate probing, and dead-zone mapping under extreme stress—round out the framework. On the Crop Recommendation Dataset (2,200 samples, 22 classes), the ensemble backbone achieves 99.1% accuracy under nominal conditions, yet a +6°C warming combined with a 35% rainfall reduction flips the recommendation for approximately 62% of test instances. That figure, invisible to any accuracy metric, is precisely what this work is designed to surface.

Index Terms—crop recommendation; climate robustness; counterfactual explanations; transition matrix; knowledge distillation; SHAP; ensemble learning

I. INTRODUCTION

Machine learning has become a reliable tool for crop recommendation. Given a handful of soil and climate measurements—nitrogen (N), phosphorous (P), potassium (K), temperature, humidity, pH, and rainfall—standard classifiers map input conditions to the most suitable crop from a fixed catalogue with impressive consistency. Random Forest, Support Vector Machines, and ensemble methods regularly surpass 98% accuracy on curated benchmarks [1], [2], a level of performance that has remained largely stable for several years. That stability, however, conceals a structural weakness that has gone largely unaddressed: every one of

these evaluations assumes that the conditions at deployment time will closely resemble those in the training data.

That assumption is increasingly difficult to justify. Climate change is driving measurable, sustained shifts in regional temperature and precipitation [3], and intra-seasonal variability means that even within a single growing year, actual conditions can deviate substantially from the historical norms a model was trained on. A system evaluated purely on held-out accuracy—drawn from the same distribution as training—gives no signal about whether its recommendations would survive a 2°C warming or a moderate rainfall deficit. If it recommends rice and a farmer commits to that crop, only to have the model’s own logic favor millet under the actual season’s conditions, the system has effectively withheld information that mattered.

The gap between point prediction and meaningful decision support is what motivates this work. Rather than chasing another fraction of a percentage point in accuracy, the focus here is on what happens to an already-accurate model’s outputs when the environment shifts—and on how to communicate that uncertainty to the people who depend on the recommendation. The backbone classifier used is deliberately standard; the contribution lies entirely in the analytical layer constructed around it.

The system architecture is illustrated in Fig. 1, which shows how raw soil and climate inputs flow through the ensemble backbone into the four novel decision-support modules.

Concretely, the work makes four contributions:

- 1) A parameterized climate perturbation module that stress-tests any trained crop classifier across a range of environmental scenarios, yielding a quantitative instability rate that accuracy scores cannot capture.
- 2) A transition matrix analysis that resolves crop-level vulnerability—revealing not just *how many* recommendations change under climate stress, but precisely *which* crops shift and toward *which* substitutes.
- 3) A directed heuristic counterfactual engine that, for a given field and a desired alternative crop, computes the minimal set of controllable soil amendments needed to cross the model’s decision boundary.
- 4) A risk-aware portfolio recommender that replaces the single predicted label with a probability-weighted shortlist of top candidate crops—a more honest output

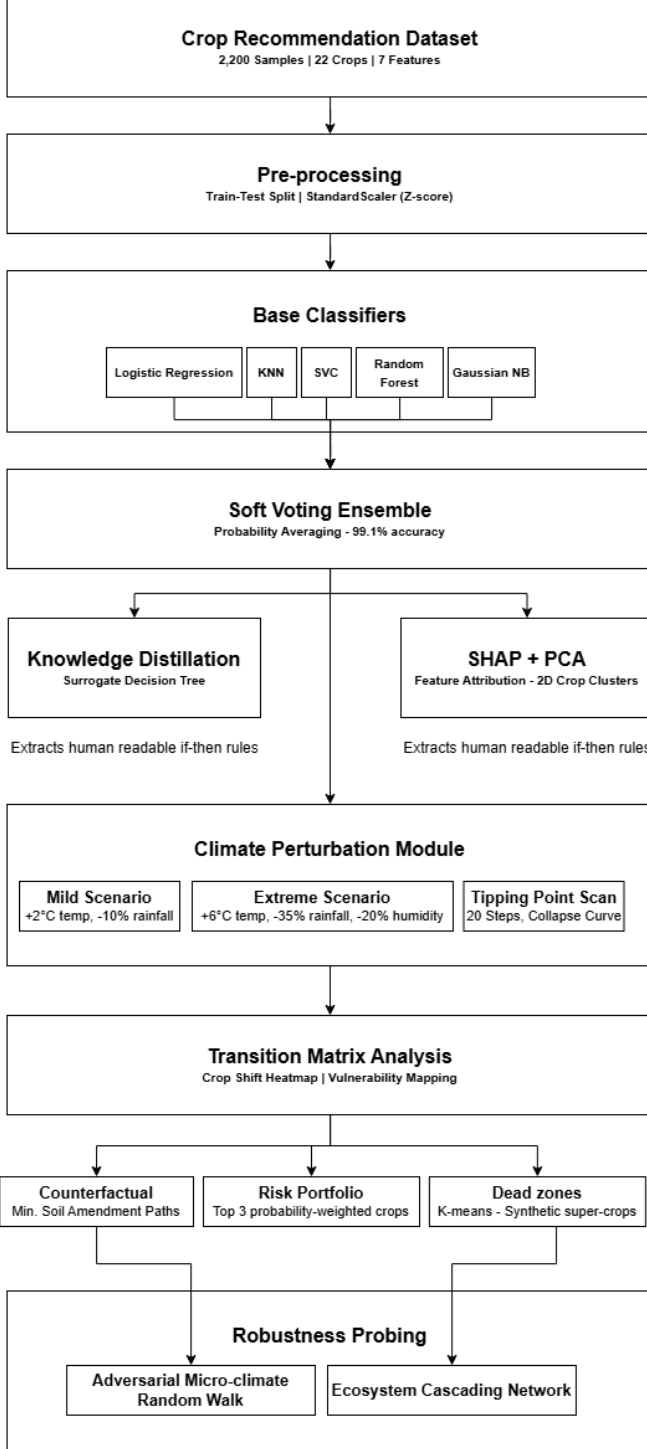


Fig. 1: End-to-end architecture of the proposed climate-aware crop decision support framework. Soil and climate inputs are processed by a soft-voting ensemble backbone, whose outputs feed four novel modules: a climate perturbation evaluator, a transition matrix analyser, a counterfactual recommendation engine, and a risk-aware portfolio recommender.

when model confidence is genuinely distributed across multiple viable options.

The paper proceeds as follows. Section II situates the work within the existing literature. Section III develops the framework in full. Section IV covers experimental setup. Section V reports and interprets findings. Sections VI and VII discuss practical implications and conclusions, and Section VIII identifies directions for follow-on work.

II. BACKGROUND AND RELATED WORK

A. ML-Based Crop Recommendation

The literature on ML-driven crop recommendation is broad and, at this point, largely convergent on accuracy. Pudumalar et al. [1] provided an early systematic comparison of Naive Bayes, Decision Trees, and Random Forest on soil-climate feature sets, demonstrating that tree-based methods consistently outperform linear alternatives. Doshi et al. [2] later extended the comparison to include deep learning variants, though the accuracy gains over Random Forest were modest on standard benchmarks. The dataset used in the present study has itself become something of a de facto testbed across this literature, with reported accuracies clustered tightly between 95 and 99%. The practical implication is that standard benchmark accuracy is effectively saturated—incremental improvements in that direction add little useful information. The more productive question is what such a model can *do* for a farmer or a policy planner beyond producing a single label.

B. Explainability in Agricultural Systems

A parallel thread of research has focused on making crop models interpretable. Lundberg and Lee’s SHAP framework [4], grounded in cooperative game theory, has been widely adopted as a principled method for attributing predictions to individual features. LIME [5] serves a similar function through local linear approximation. Surrogate tree methods [6]—training a shallow, human-readable decision tree to mimic a complex model—offer a complementary approach that trades some fidelity for immediate interpretability. All of these techniques, however, operate in a static regime: they explain *what* the model decided for a given input without addressing whether that decision would hold under a different set of conditions. The explanatory apparatus does not travel well to the robustness question.

C. Climate Robustness and Counterfactual Reasoning

Counterfactual explanation—formalised by Wachter et al. [7] as an optimisation problem over the input space—asks a fundamentally different question: given the current output, what is the smallest input change that would produce a different one? Applied to agriculture, this translates naturally into soil-amendment advice. The approach has found substantial traction in credit scoring and clinical decision support but remains underused in crop systems. Climate robustness evaluation—testing how a model’s outputs respond to systematic input perturbations—is

even less developed. The agronomic modelling community has addressed related questions through physics-based crop simulators such as DSSAT and APSIM [3], but these tools are not designed to interrogate trained ML models, and bridging the two paradigms is a largely open problem.

D. The Unaddressed Gap

Taken together, the literature has produced models that predict well and, more recently, models that explain their predictions well. What it has not produced is a framework that asks whether those predictions remain reliable under the kind of environmental stress that characterises real-world agricultural conditions now and in the near future. None of the reviewed systems exposes a trained model to parameterized climate scenarios, quantifies the resulting recommendation instability, or provides actionable guidance when stability breaks down. Those are the gaps this work fills.

III. PROPOSED FRAMEWORK AND METHODOLOGY

The framework separates cleanly into two layers. The first—a soft-voting ensemble of five classifiers—is a standard construction described in Section III-A; it is included for completeness and to establish a high-accuracy baseline, but it is not the contribution. The second layer, detailed in Sections III-B through III-E, contains the four novel components that convert the backbone into an adaptive decision support system.

A. Standard Backbone

Five classifiers form the predictive core: Logistic Regression, K-Nearest Neighbours ($k = 5$), a Support Vector Classifier with an RBF kernel, Random Forest, and Gaussian Naive Bayes. All seven features are standardised to zero mean and unit variance using statistics computed on the training split alone—no leakage to the test partition. The five classifiers are combined through soft voting, which averages their class probability vectors rather than taking a plurality vote:

$$\hat{P}(y = c \mid \mathbf{x}) = \frac{1}{N} \sum_{i=1}^N P_i(y = c \mid \mathbf{x}), \quad (1)$$

where $N = 5$ and $P_i(y = c \mid \mathbf{x})$ is the posterior probability assigned to class c by the i -th model. This is the `VotingClassifier(voting='soft')` available in `scikit-learn`.

Before model training, the pairwise Pearson correlations among all seven input features were examined. Fig. 2 presents the resulting correlation matrix. The most prominent relationship is a strong positive association between Phosphorous (P) and Potassium (K) ($r = 0.74$), suggesting that these two nutrients co-vary systematically across the crop environments in this dataset. All other inter-feature correlations are weak ($|r| < 0.25$), which confirms that the feature set is largely non-redundant and that most variables contribute independent information to the classifier.

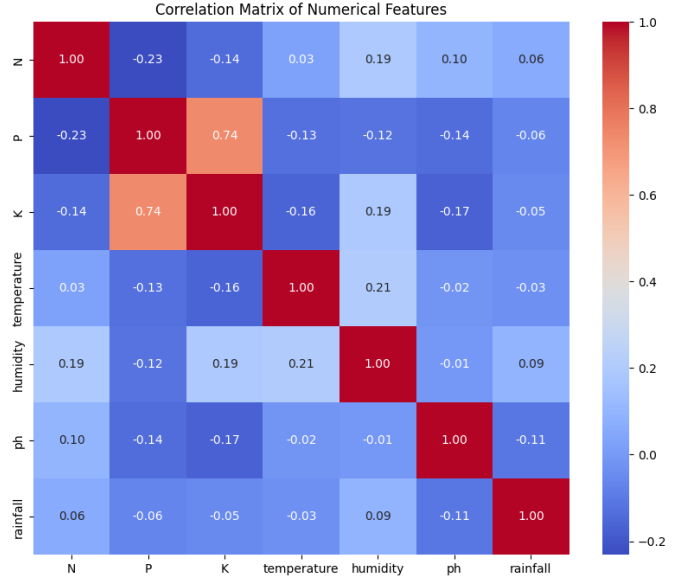


Fig. 2: Pearson correlation matrix across all seven input features. The pronounced positive correlation between Phosphorous and Potassium ($r = 0.74$) is the only strong inter-feature dependency; the remaining pairs are weakly correlated, indicating that each feature contributes largely independent information to the model.

Two interpretability tools are layered on top of the ensemble. First, a surrogate decision tree (`max_depth = 5`) is trained on the ensemble’s pseudo-labels over the full training set, implementing a lightweight form of knowledge distillation [6] that produces human-readable if-then rules. The structure of this tree is shown in Fig. 3: rainfall governs the root split, partitioning instances into low- and high-rainfall regimes, after which potassium, humidity, and phosphorous progressively resolve finer crop distinctions toward the leaves. Second, SHAP’s `TreeExplainer` is applied to the Random Forest to generate a global feature attribution ranking.

B. Climate Perturbation Module

The central analytical component takes a trained classifier f and a test partition $\mathbf{X}_{\text{test}}$, then constructs a perturbed counterpart $\mathbf{X}_{\text{test}}(\delta)$ by applying structured offsets to the two continuous climate variables:

$$\mathbf{X}_{\text{test}}(\delta) : \begin{cases} \text{temperature} \leftarrow \text{temperature} + \Delta T \\ \text{rainfall} \leftarrow \text{rainfall} \times (1 - \Delta r) \end{cases} \quad (2)$$

The instability rate—the fraction of test instances where the classifier’s output changes after perturbation—quantifies how sensitive the recommendation system is to that particular environmental shift:

$$\text{Instability}(\delta) = \frac{|\{i : f(\mathbf{x}_i) \neq f(\mathbf{x}_i(\delta))\}|}{|\mathbf{X}_{\text{test}}|}. \quad (3)$$

Surrogate Decision Tree Visualization

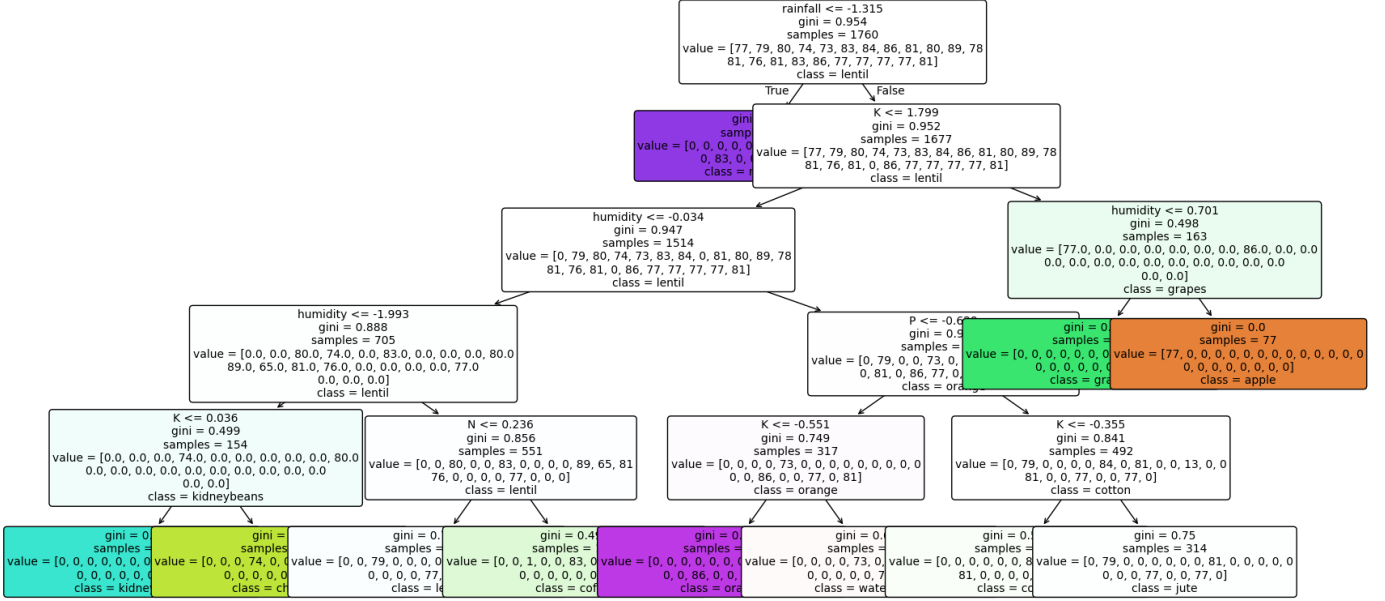


Fig. 3: Surrogate decision tree (max_depth = 5) distilled from the soft-voting ensemble. Rainfall serves as the root split feature, with subsequent nodes using K, humidity, and P to progressively disambiguate crop classes. Leaf colours correspond to the final predicted class; Gini impurity values quantify residual uncertainty at each node.

Beyond evaluating fixed scenarios, the module runs a sweep: ΔT is increased from 0 to $+10^\circ\text{C}$ in 20 half-degree steps, with rainfall decaying by a further 5% at each step. The resulting *agricultural collapse curve* traces how the instability rate grows with compounding environmental stress and locates the temperature at which it crosses 50%—a threshold termed the *50% collapse point*. This curve converts an otherwise opaque sequence of model queries into a single, interpretable robustness profile.

C. Transition Matrix Analysis

A single aggregate instability rate conceals important class-level information. Two scenarios with identical instability rates can differ dramatically in *which* crops are affected and *where* displaced recommendations land. The transition matrix $\mathbf{T}$ resolves this by cross-tabulating original and perturbed predictions:

$$T[c, c'] = |\{i : f(\mathbf{x}_i) = c \text{ and } f(\mathbf{x}_i(\delta)) = c'\}|. \quad (4)$$

Strong diagonal entries signal prediction-stable crops; off-diagonal entries identify the specific substitution patterns induced by climate stress. A crop is formally designated *climate-vulnerable* when its diagonal retention falls below 70%:

$$\frac{T[c, c]}{\sum_{c'} T[c, c']} < 0.7. \quad (5)$$

This per-class vulnerability measure gives policymakers a targeted, evidence-based picture of which crops need priority attention in climate adaptation planning.

D. Counterfactual Recommendation Engine

When a farmer's field is currently predicted as crop c_{orig} but a different crop c_{target} is desired—perhaps because c_{orig} is projected to become vulnerable under climate change—the counterfactual engine identifies the smallest feasible soil-amendment that would redirect the model's recommendation. The underlying problem is:

$$\min_{\mathbf{x}'} d(\mathbf{x}, \mathbf{x}') \text{ s.t. } f(\mathbf{x}') = c_{\text{target}}, \quad \mathbf{x}'_S \in \mathcal{C}_S, \quad (6)$$

where $S = \{N, P, K, \text{pH}\}$ are the features a farmer can realistically amend, $\mathcal{C}_S$ enforces agronomically plausible ranges, and $d(\cdot, \cdot)$ is the ℓ_1 distance over those features. Temperature, humidity, and rainfall are held fixed—they are outside the farmer's control.

Because Random Forest does not expose a differentiable loss surface, gradient descent is not applicable. The engine instead runs a directed heuristic: at each iteration, every controllable feature nudges by a small step α toward the training-set mean of c_{target} , and the updated feature vector is re-queried. Iteration stops as soon as the classifier returns c_{target} , or after a fixed budget of steps. The output—something like “raise N by 18 kg/ha, raise P by 21 kg/ha, raise K by 11 kg/ha, lower pH by 0.3 units”—is immediately actionable by an agricultural extension officer. The heuristic does not

guarantee the globally minimal perturbation, a limitation addressed in Section VIII.

E. Risk-Aware Crop Portfolio Recommender

A single predicted label is an appropriate output when the model is confident. Under environmental stress, however, that confidence often spreads across several plausible crops—and forcing a single-label output discards information the farmer needs. The portfolio component addresses this by normalising the Random Forest’s top-three posterior probabilities into an explicit allocation:

$$\pi_k = \frac{P_{\text{RF}}(y = c_k \mid \mathbf{x}(\delta))}{\sum_{j=1}^3 P_{\text{RF}}(y = c_j \mid \mathbf{x}(\delta))}, \quad (7)$$

where $\{c_1, c_2, c_3\}$ are ranked by descending probability. The result—for example, millet (48%), sorghum (31%), barley (21%)—communicates distributional uncertainty in a form that directly informs planting decisions, much as a financial portfolio distributes exposure when the future is uncertain.

Complementing the portfolio, a *dead-zone detector* flags instances where $\max_k P_{\text{RF}} < \tau = 0.5$: conditions so severe that no known crop commands majority model confidence. K-Means ($k = 3$) clustering over these flagged instances in the $\{N, P, K, \text{pH}\}$ subspace produces three centroid profiles—soil-nutrient blueprints for hypothetical stress-resilient varieties that no existing crop in the training catalogue can currently fill.

IV. IMPLEMENTATION DETAILS

A. Dataset

All experiments use the publicly available Crop Recommendation Dataset [1]: 2,200 instances spanning 22 crop classes at exactly 100 samples per class, described by the seven features listed earlier. The perfect class balance removes class-imbalance as a confounding factor in the baseline evaluation, though it is also a known limitation of the benchmark’s synthetic construction. An 80/20 stratified split (`random_state = 42`) yields 1,760 training and 440 test instances. A single split was used throughout; the implications of this choice are discussed in Section VIII.

To understand how crops occupy the seven-dimensional feature space, PCA was used to project the full scaled dataset onto two principal components. The resulting scatter plot, shown in Fig. 4, reveals a clear three-region structure. Chickpea forms a compact, isolated cluster at the lower end of the plot—separated from all other crops by a substantial margin in PC2. A second distinct group comprising muskmelon, apple, watermelon, and grapes occupies the high-PC1 region on the right. The large central mass contains the remaining pulses, cereals, and fruits, with considerable overlap—a zone of genuine classification ambiguity that the robustness framework is particularly suited to interrogate.

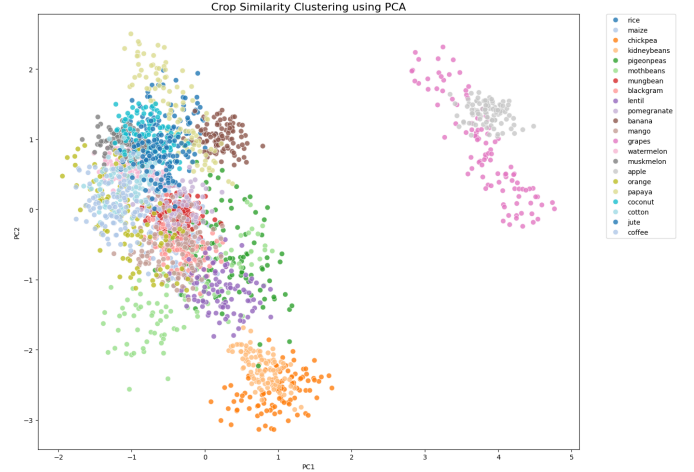


Fig. 4: Two-component PCA projection of the full scaled dataset, with each point coloured by crop label. Chickpea (orange) and the muskmelon-apple-watermelon-grapes cluster (upper right) are well-separated in this space. The dense central region contains a mix of pulses and cereals whose soil-climate profiles substantially overlap, representing the area of highest classification ambiguity.

B. Tools and Libraries

The entire pipeline runs in Python 3.10. Ensemble training, scaling, and K-Means use scikit-learn 1.3; SHAP 0.43 handles feature attribution; NetworkX 3.1 constructs the ecosystem correlation graph; and NumPy, Pandas, Matplotlib, and Seaborn cover numerical computation and visualization throughout. No GPU hardware was required at any stage.

C. Hyperparameters

All five classifiers run with scikit-learn defaults except where noted. Logistic Regression uses `max_iter = 1000` and the `lbfgs` solver. KNN uses five neighbours. The SVC uses an RBF kernel with `probability=True` to enable soft voting. Random Forest uses 100 trees with `max_features='sqrt'`. Gaussian Naive Bayes uses default class priors. No hyperparameter search was performed—the backbone is not the locus of the contribution, so optimising it would be beside the point.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Baseline Performance

Table I records the test-set accuracy of each component classifier alongside the soft-voting ensemble. Random Forest and Gaussian Naive Bayes both reach approximately 99.3%, with the ensemble settling at 99.1%—marginally *lower* than the best individual model. This is not surprising: on a dataset where several classifiers have already learned near-perfect boundaries, averaging their probability outputs offers no meaningful benefit and can slightly dilute the sharpest model’s signal. In any production deployment of this backbone, Random Forest alone would be the rational choice.

TABLE I: Baseline Classifier Accuracy (test set, $n = 440$)

Model	Accuracy (%)	Notes
Logistic Regression	~95.5	Linear boundary; weakest
Gaussian Naive Bayes	~99.3	Violated indep. assumption
KNN ($k = 5$)	~97.5	Sensitive to scale
SVC (RBF)	~98.6	Strong non-linear boundary
Random Forest	~99.3	Best individual model
Soft-Voting Ensemble	~99.1	Combines all five

B. SHAP Feature Attribution

Global feature attribution across all 22 classes, shown in Fig. 5, confirms that humidity and K dominate the model’s average decision signal, with rainfall and N contributing at a similar level. Temperature and pH rank lowest globally, though this masks meaningful class-specific variation—a crop like coffee, for instance, relies heavily on a specific temperature band even if that feature ranks low in the aggregate. The per-class decomposition (stacked bars) makes these intra-feature contrasts directly readable.

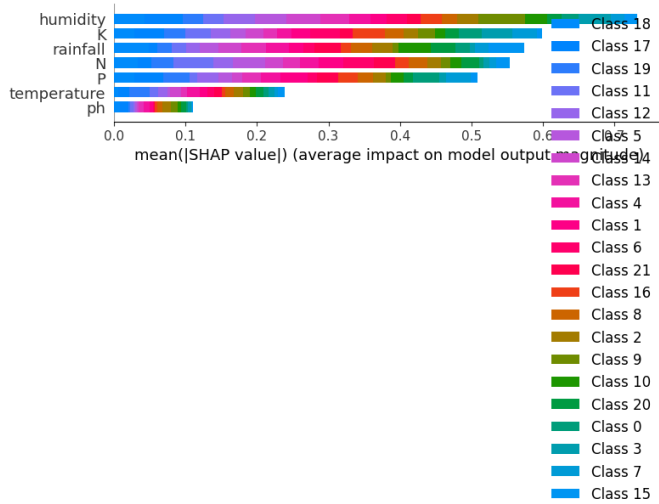


Fig. 5: SHAP global feature importance (mean |SHAP value|) for the Random Forest model, decomposed by crop class. Humidity and K contribute most on average, while temperature and pH rank lowest globally yet show meaningful variation across individual crop classes in the stacked per-class breakdown.

C. Moderate Climate Scenario

Under a modest perturbation—temperature up by 2°C, rainfall down by 10%—approximately 17% of test-set recommendations change. That fraction, roughly one in six fields, is non-negligible; a farmer following the original recommendation in any of those cases would be acting on advice that the model’s own logic would revise under the actual seasonal conditions. The transition matrix for this scenario, Fig. 6, shows that most crops remain firmly on the diagonal. The exceptions are informative: rice shifts toward

jute in six instances, and a small proportion of blackgram predictions migrate to lentil. Both are rainfall-sensitive transitions—exactly what the 10% rainfall reduction would be expected to trigger.

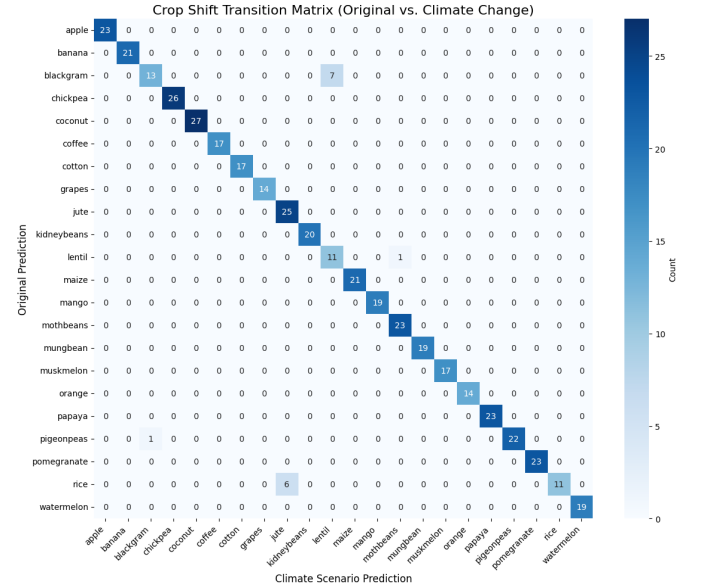


Fig. 6: Crop shift transition matrix under the moderate climate scenario ($\Delta T = +2^\circ\text{C}$, rainfall -10%). The strong diagonal reflects overall prediction stability, while off-diagonal entries—most notably rice shifting to jute and blackgram to lentil—identify the crops whose narrow rainfall tolerances make them first to be affected by even moderate perturbation.

D. Extreme Climate Scenario

Scaling the perturbation to $+6^\circ\text{C}$, -35% rainfall, and -20% humidity produces a qualitatively different picture. The extreme transition matrix in Fig. 7 shows widespread off-diagonal activity. Rice migrates entirely to coffee; coconut fragments across several substitute classes including maize and pigeonpeas; lentil splits between maize and mungbean; and mungbean redirects substantially toward kidneybeans. Against this backdrop, a set of crops—apple, banana, chickpea, grapes, and mothbeans—maintain strong diagonal retention, emerging as the most climate-robust options within the model’s knowledge. The red colour scale of this matrix, deliberately contrasted with the blue used for the moderate scenario, makes the severity of the shift visually immediate.

The 62% aggregate instability under these extreme conditions cannot be read from any accuracy score. A system reporting 99.1% accuracy would give no hint of how profoundly its recommendations reorganise under plausible future climate conditions.

E. Agricultural Collapse Curve

The iterative tipping-point simulation traces how the aggregate failure rate evolves as temperature and rainfall deteriorate together. The resulting collapse curve, shown in

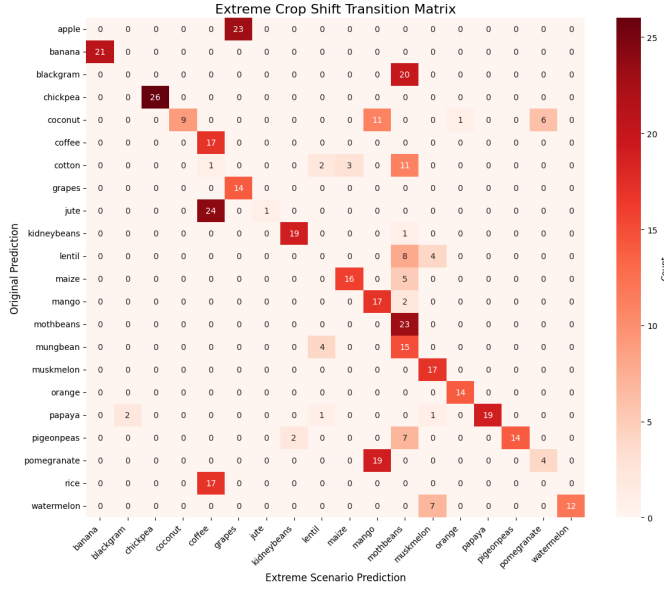


Fig. 7: Extreme crop shift transition matrix ($\Delta T = +6^\circ\text{C}$, rainfall -35% , humidity -20%). Off-diagonal counts are substantially higher than in the moderate scenario. Rice, coconut, lentil, and mungbean exhibit the greatest instability; apple, banana, chickpea, and mothbeans retain strong diagonal entries, marking them as the most climate-robust crops in this dataset.

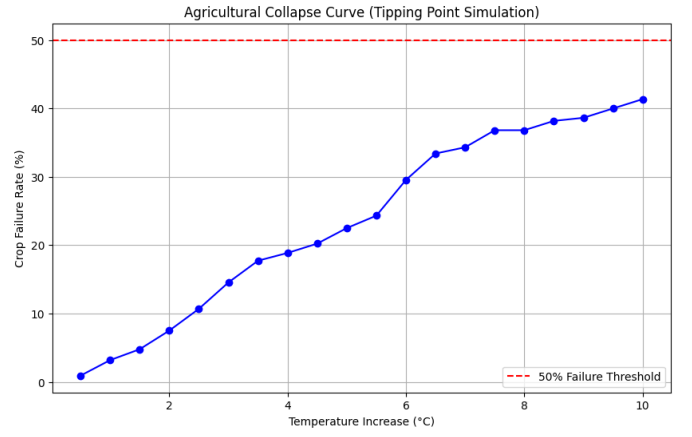


Fig. 8: Agricultural collapse curve: the percentage of test instances receiving a changed crop recommendation as a function of cumulative temperature increase (with compound 5% rainfall decay per step). The 50% instability threshold is approached but not reached within the $+10^\circ\text{C}$ range, indicating that the model continues to recommend some crop for most inputs even under severe stress—though not necessarily the agronomically appropriate one.

Fig. 8, rises monotonically from roughly 1% instability at $+0.5^\circ\text{C}$ to 41% at $+10^\circ\text{C}$. The 50% threshold (dashed red line) is not crossed within the simulated range—a finding with an important interpretation. The model does not run out of predictions; it substitutes one crop for another. Whether those substitutions are agronomically appropriate for the shifting climate is a question the model cannot answer on its own, which is precisely why the dead-zone and portfolio components exist to flag conditions where confidence is genuinely distributed or absent.

F. Dead Zones and Portfolio Recommendations

Under extreme conditions, a portion of test instances produced a maximum class probability below 0.5—cases where the model is genuinely agnostic among several crops. K-Means clustering of these dead-zone instances in the $\{N, P, K, \text{pH}\}$ space yielded three centroids with distinct nutrient profiles, providing a data-driven target specification for stress-resilient crop development. For one representative dead-zone instance, the portfolio recommender output was: millet (48%), sorghum (31%), barley (21%)—a richer and more honest recommendation product than forcing a single label.

G. Counterfactual Example

A concrete illustration: a test instance originally mapped to rice ($N = 82$, $P = 46$, $K = 42$, $\text{pH} = 6.2$, temperature = 24°C , humidity = 82%, rainfall = 210 mm) was redirected

to jute by the counterfactual engine through the following amendments: $N + 18$ kg/ha, $P + 21$ kg/ha, $K + 11$ kg/ha, $\text{pH} - 0.3$. Each of these is a standard soil-management intervention with a known cost, making the output directly usable by field agronomists.

H. Ecosystem Feature Dependency Network

To understand why certain climate perturbations produce cascading crop shifts, the correlation structure of the feature space was represented as a network graph, shown in Fig. 9. K emerges as the highest-degree hub node, connected to P (the strongest individual edge, $r = 0.74$), humidity, temperature, and pH. This hub structure means that environmental changes which co-shift moisture and nutrient availability propagate through the K node into multiple prediction pathways simultaneously. Rainfall and pH sit at the network periphery—consistent with their lower SHAP attribution and their relative independence from other features.

VI. APPLICATIONS

A. Farmer Decision Support

The framework is designed to speak at multiple levels simultaneously. At the field level, a farmer or extension officer receives three outputs: a point recommendation under current conditions, a portfolio recommendation weighted by model confidence under a projected climate scenario, and—if they wish to grow a specific crop—a set of concrete soil amendments that would move their field into that crop’s decision region. The last output is particularly well-suited to programs like India’s Soil Health Card scheme, where soil-amendment guidance is already an established delivery mechanism.

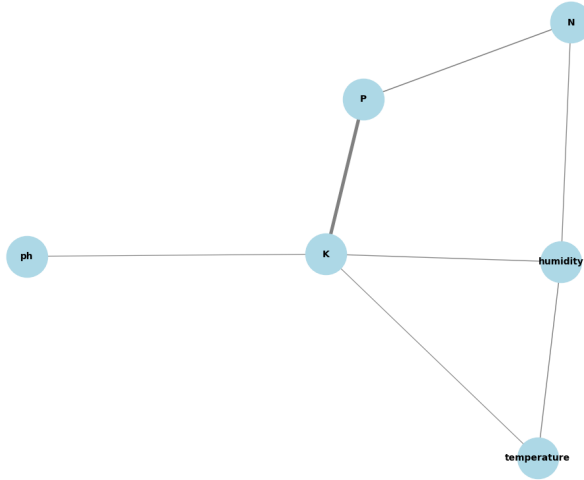


Fig. 9: Ecosystem cascading network of soil and climate features, constructed from pairwise Pearson correlations exceeding $|\rho| > 0.15$. Edge width is proportional to correlation strength. K acts as the central hub, linked to P, humidity, temperature, and pH; rainfall and pH are peripheral nodes with few strong inter-feature connections—a pattern consistent with their lower global SHAP scores.

B. Policy Planning

The transition matrices and collapse curves scale naturally to district or regional analysis. By coupling the perturbation module to CMIP6 or ERA5 projections specific to a target geography, planners could generate maps showing, crop by crop, where recommendation instability is projected to exceed acceptable thresholds under RCP 4.5 versus RCP 8.5 scenarios. Such maps would give concrete, quantitative grounding to crop diversification subsidy decisions and climate adaptation fund allocations.

C. Crop Breeding and Biotechnology

The dead-zone centroids produced by K-Means clustering define a soil-nutrient envelope that no currently catalogued crop occupies with majority model confidence under severe stress. That envelope is, in effect, a data-derived phenotype specification: a description of the conditions that a stress-tolerant variety would need to colonise in order to fill the gap. Biotechnology programs working on drought- and heat-tolerant varieties could use these centroids as quantitative targets for trait engineering.

VII. CONCLUSIONS

The central argument of this paper is simple: a crop recommendation system that achieves 99.1% accuracy on a held-out test set has not demonstrated that it is trustworthy for real-world agricultural planning. It has demonstrated that it memorises the statistical patterns in one particular dataset. Whether those patterns still hold under a different climate is a

separate question—and it is the question that actually matters to the farmer who will act on the recommendation.

The framework developed here answers that question through four components that together span the distance from static accuracy to dynamic, climate-aware decision support. The moderate scenario reveals that roughly one in six recommendations changes under a 2°C warming; the extreme scenario flips nearly two-thirds. The transition matrices make these shifts legible at the crop level. The collapse curve locates the tipping point. The portfolio recommender communicates residual uncertainty honestly. And the counterfactual engine converts the model’s decision boundary into an actionable soil-amendment prescription.

None of these components requires a new algorithm. The contribution is architectural: a way of wrapping and interrogating an existing classifier that extracts information the classifier has never been asked to provide.

VIII. FUTURE WORK

Several avenues for meaningful extension are apparent. The most pressing is data: replacing the synthetic benchmark with real soil health card data, field-survey records, or remote-sensing-derived inputs would test whether the framework’s findings generalise beyond a carefully curated laboratory setting. Closely related, the perturbation module currently uses hand-tuned ΔT and Δr values; integrating it with CMIP6 or ERA5 reanalysis data for a specific region would ground the collapse curves in actual projected climate trajectories.

On the counterfactual side, the current heuristic search has no guarantee of minimality. Replacing it with DiCE [8] or a manifold-constrained variant of Wachter’s optimisation [7] would both improve the quality of the amendments suggested and ensure that the counterfactuals lie within the realistic data distribution rather than at arbitrary points in feature space.

Further ahead, a reinforcement learning formulation—where the agent learns a planting policy that maximises long-term yield under dynamically evolving climate trajectories—would move the system from one-shot recommendation to sequential adaptive planning. Coupling that with multi-objective optimisation that balances agronomic viability against economic yield, water consumption, and market price risk would bring the decision support model considerably closer to the complexity of real farming decisions.

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